

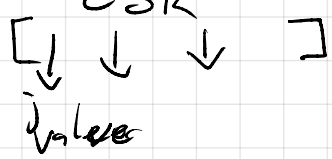
COO

i

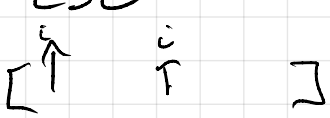
j

values

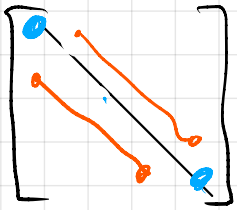
CSR



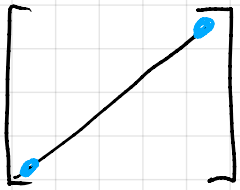
CSC



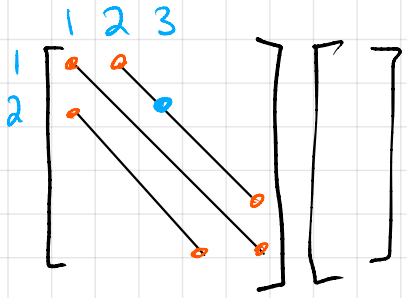
Structure exploitation



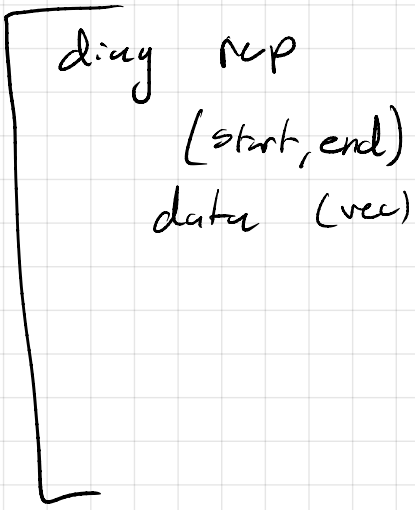
$i = j$



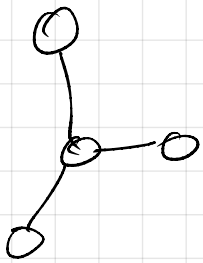
$i = N - j$

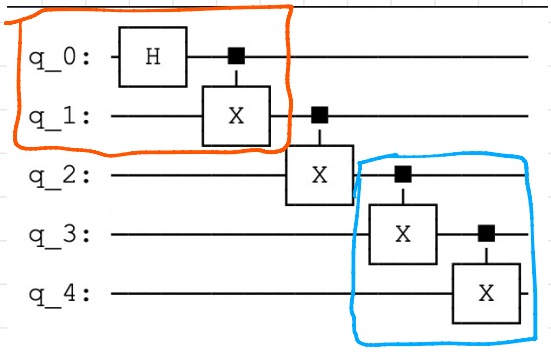


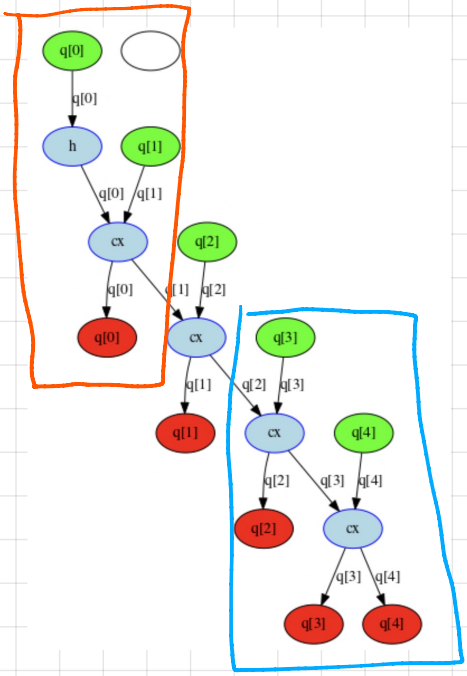
[2, 3]



$H \otimes H$







$$(CNOT)(H \otimes I)(q_0 \otimes q_1)$$

GOAL: Memory Efficiency
(not contraction choices)

$$[(CNOT)[(H \otimes I)(H \otimes I)(H \otimes I)(H \otimes I)](q_0 \otimes q_1)]$$

New struct:

start i, j
end k, l

data : [, , , , , , ,]

complex "integers"

a + bi w/ a, b ∈ ℤ

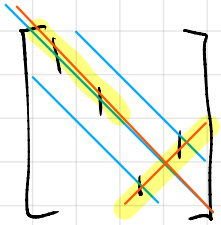
a, b ∈ {-1, 0, 1}

Most existing pkgs use numpy

Dense ⇒ ~~struct?~~ ↳ Class?

$$\begin{bmatrix} 1 & 2 & 0 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \Rightarrow$$

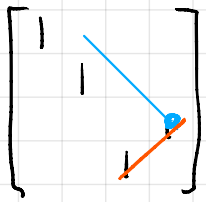
Assume principal diagonals

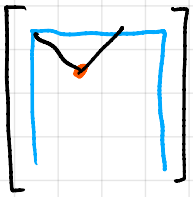


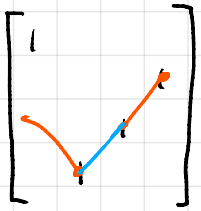
$H H^*$ do we need constants?

$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

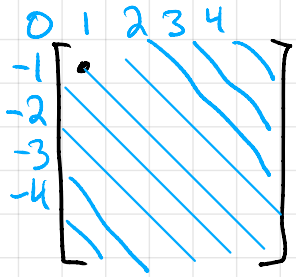
$\sqrt{1+1} = \sqrt{2}$

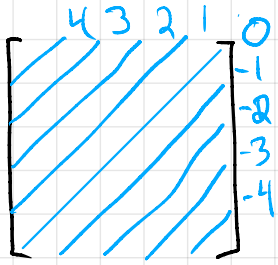






10/28





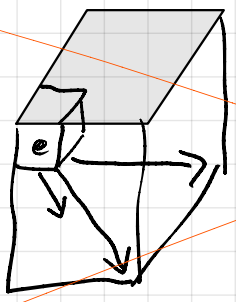
$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}$$

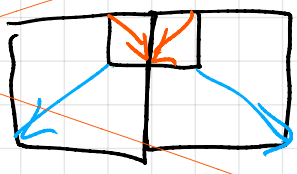
only use principal diags

(sparsity ratio criteria?)

$$\begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix}$$

consider diagonal sparsity? later





Stick to 2D! (don't need more for QC)

TODOs:
- Matrix (complexes) (srikar)
- Vector class (Blake) (and cleanup general struct)
- start kernels (M·V (Blake), M·M (srikar), M⊗M later...)
- finish Dense/sparse conversions (srikar)

M·V $\begin{bmatrix} 1 & \\ & \end{bmatrix}\begin{bmatrix} 2 \\ 1 \end{bmatrix}$

M·M $\begin{bmatrix} & \end{bmatrix}\begin{bmatrix} & \end{bmatrix}$

Maybe?

$a \cdot b = a^T b$

$\langle a | b \rangle$

Shoot to have these done by 11/1

$A \otimes B \begin{bmatrix} A_{11}[B] & A_{12}[B] \\ A_{21}[B] & A_{22}[B] \end{bmatrix}$

11/4 Mat - Mat mult

A B C

$\begin{bmatrix} \text{——} \end{bmatrix}\begin{bmatrix} \mid & \end{bmatrix} = \begin{bmatrix} \cdot & \end{bmatrix}$

$C_{ij} = \sum_k A_{ik} B_{kj}$

IF we can access element (BUT doesn't

$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}\begin{bmatrix} 0 & & & \\ & k_1 & & \\ & & \ddots & \\ & & & k_{n-1} \end{bmatrix} = \begin{bmatrix} \mid \\ 0 \\ \mid \end{bmatrix}$

$\begin{bmatrix} \mid & [A_s] \\ a_{n1} & \text{——} \end{bmatrix}\begin{bmatrix} \mid & [\,|k\,]] \\ 0 & \\ \mid & \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} \mid & [A_s k] \\ 0 & \\ \mid & \text{——} \end{bmatrix}$

$A_s k = \begin{bmatrix} 1 & & & \\ k_1 & a_{12} & & \\ & 1 & & \\ & a_{n-1,2} & & \\ & 1 & & \end{bmatrix} \quad k_2 \begin{bmatrix} a_{13} \\ a_{23} \\ \vdots \\ a_{n-1,2} \\ \vdots \end{bmatrix} \quad \cdots \quad \begin{bmatrix} \\ \\ \\ \\ \\ \end{bmatrix}$

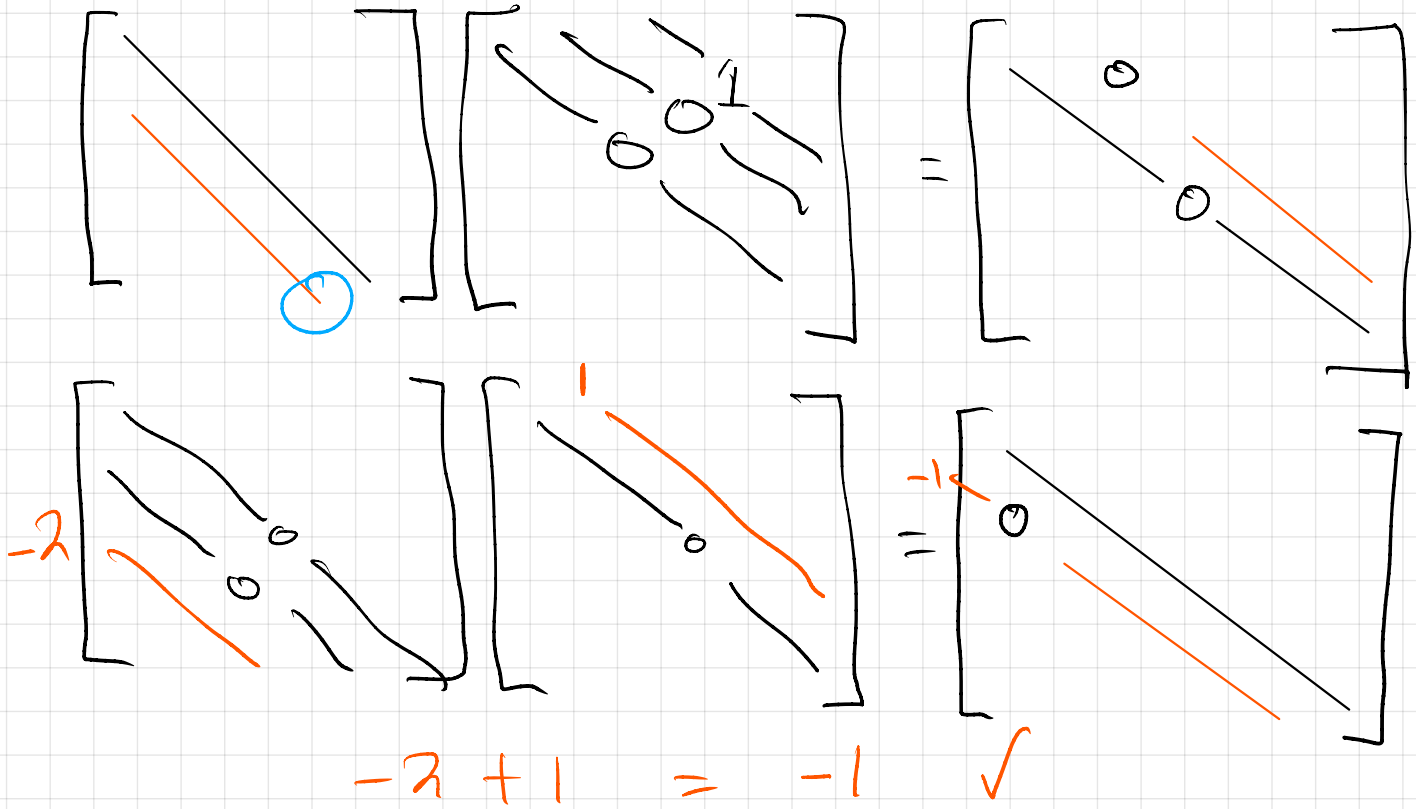
$\in \mathbb{C}^{n-1 \times n-1}$

$\begin{bmatrix} a_{11} & & \\ & a_{22} & \\ & & a_{33} \\ & & \ddots \end{bmatrix} \begin{bmatrix} b_{12} & & \\ & b_{23} & \\ & & \ddots \end{bmatrix} = \begin{bmatrix} a_{11}b_{12} & & \\ & a_{22}b_{23} & \\ & & \ddots \end{bmatrix}$

$\begin{bmatrix} a_{11} & 0 \\ 0 & A \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & B \\ 1 & 0 \end{bmatrix} \qquad \begin{bmatrix} \end{bmatrix} \begin{bmatrix} B \end{bmatrix}$

$\begin{bmatrix} 0 & a_{12} & & \\ & a_{23} & & \\ & & \ddots & \\ & & & 0 \end{bmatrix} \begin{bmatrix} 0 & b_{12} & & \\ & b_{23} & & \\ & & \ddots & \\ & & & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 2 & \\ 0 & 0 & a_{12}b_{23} & \\ & & a_{23}b_{34} & \\ & & & \ddots \end{bmatrix}$

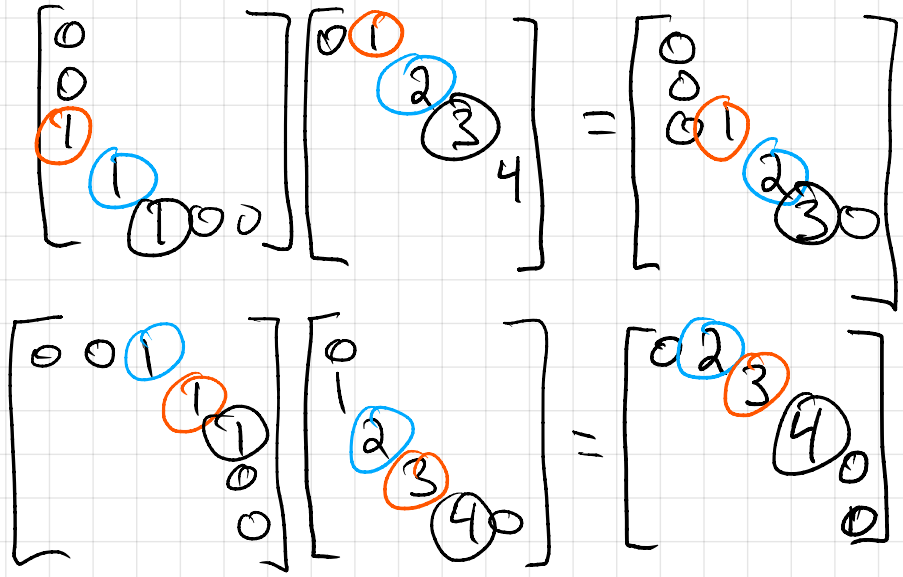
$\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} \begin{bmatrix} 0 & & \\ & & \\ & & 0 \end{bmatrix} = \begin{bmatrix} 0 & & \\ & & \\ & & 0 \end{bmatrix}$

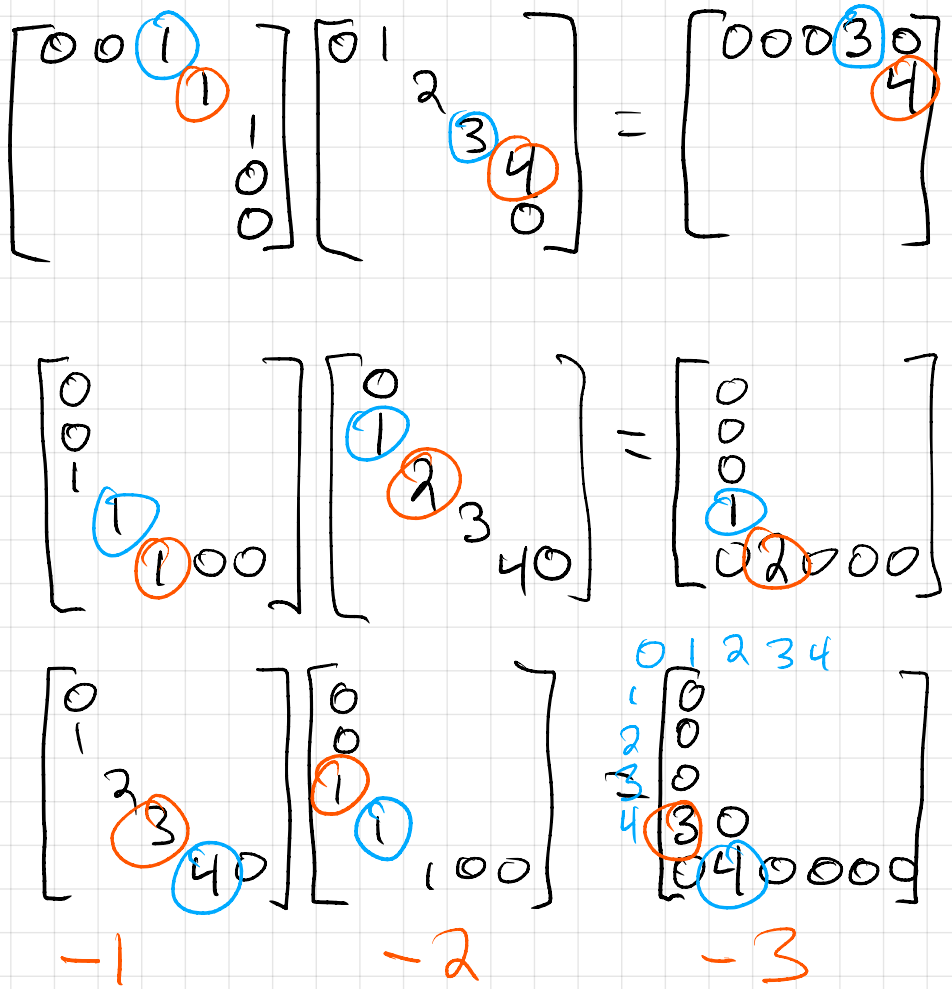


-2 : index [ ]

storage [ ]

-2 : [-1, 0, 1]





-1 -2 -3

Shift 2 $\Rightarrow$ losing first 2 on -1

$$A \cdot B = (A_0 + A_1 + A_{-1} + \cdots)(B_0 + B_{-1} + B_1 + \cdots)$$

$(2N-1)(2N-1)$

A[index] diag

B[index] diag

A[i]

B[j]

$A_0 \cdot B_n$

[nelement] [1]

$A, B \in \mathbb{C}^{N \times N}$

#diags $2N-1$

$A_0 B_0$ N

$A_0 B_1$ $N-1$

$\vdots$ $\vdots$

$A_0 B_n$ 1

$A_1 B_0 : N-1$

$A_1 B_1 : N-2$

$\vdots$

$A_1 B_{N-1} : 1$

$A_1 B_N : 0$

$\begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} \cdot & 1 \\ \cdot & \cdot \end{bmatrix} =$

	A_{-2}	A_{-1}	A_0	A_1	A_2
B_{-2}	0	N-3	N-2	N-3	0
B_{-1}	N-3	N-2	N-1	N-2	N-3
B_0	N-2	N-1	N	N-1	N-2
B_1	N-3	N-2	N-1	N-2	N-3
B_2	0	N-3	N-2	N-3	0

1, 4, 8, 12

N=3

$$\sum_{i=0}^{N-1} (4i)(N-i)$$

$$= \sum_{i=0}^{N-1} 4iN - \sum_{i=0}^{N-1} 4i^2$$

$\in \Theta(N^2)$ $\in \Theta(N^3)$

Not breaking complexity